Time: 25 Minutes **Marks: 10**

Section C1 — Heuristic-Based Move Ordering for Alpha-Beta

Modify your existing Alpha-Beta Pruning implementation so that valid moves are searched in a heuristic-based order.

Required move ordering:

For a maximizing player:
1. Generate all valid moves.
2. Simulate each move.
3. Evaluate the resulting state using your existing heuristic.
4. Sort moves from highest heuristic value to lowest.
5. Search the moves in this order.

For a minimizing player:
- Search moves from lowest heuristic value to highest.

Requirements:
1. Count the number of nodes visited using the original move ordering.
2. Count the number of nodes visited using the new heuristic-based ordering.
3. Print:
- Best Move
- Nodes Visited with Original Ordering
- Nodes Visited with New Ordering

The same search depth and evaluation function must be used in both cases.